The Global Obesity Drug Race and its Implications for Investors

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1. The global obesity drug race and its implications for investors

In 2022, 890 million adults, or 16% of the world's adults were obese according to the World Health Organization. Obesity, defined by having a body mass index greater than 30, dramatically increases the likelihood of type 2 diabetes, cardiovascular disease, hypertension, stroke, and certain cancers. It is now considered the second leading cause of preventable death globally, after smoking. The World Obesity Federation's 2023 Atlas reports that overweight and obesity could reduce the global economy by over US $4 trillion per year by 2035.

Obesity drugs, currently taking the form of GLP-1s such as Wegovy, Ozempic and Mounjaro, are hailed as the statins of the 21st century and are predicted to represent more than $100 billion in annual market potential. Breakthroughs in GLP-1 efficacy, positive trial results, and soaring demand have already sent the share prices of market leaders surging: Novo Nordisk overtook LVMH as Europe's most valuable company in 2023, while Eli Lilly's valuation doubled in just two years.

Yet, this pharmaceutical arms race is far from settled. Several biotech challengers including Amgen, Pfizer, and smaller clinical-stage firms are racing to develop oral formulations and dual or triple-pathway drugs that could surpass current GLP-1 injections in effectiveness and convenience. For investors, this means the near-term winners are likely to be the companies that successfully commercialize new therapies, scale production, and maintain patent exclusivity.

Successfully navigating the potential of obesity drugs can be a great driver of returns in the portfolio of investors. Novo Nordisk, the current market leader, has leveraged its first-mover

advantage to dominate global GLP-1 sales. Its strong clinical track record and established manufacturing infrastructure provide a moat, but slowing growth, increased competition from Eli Lilly's tirzepatide, and rising pricing pressures temper enthusiasm. Meanwhile, Eli Lilly is positioning itself as the next dominant player, with Mounjaro rapidly gaining market share and showing superior efficacy in some trials. Smaller entrants may not threaten the duopoly immediately, but their innovations, particularly oral or multi-pathway GLP-1s could erode margins and share over the medium term.

	Price per share	P/E	Dividend yield	Year range	YoY change	EPS
NVO	$55.64	14.27	3.10%	$45.05 - $121.34	-55.21%	3.95
LLY	$722.00	47.07	0.83%	$623.78 - $937.00	-20.67%	15.29

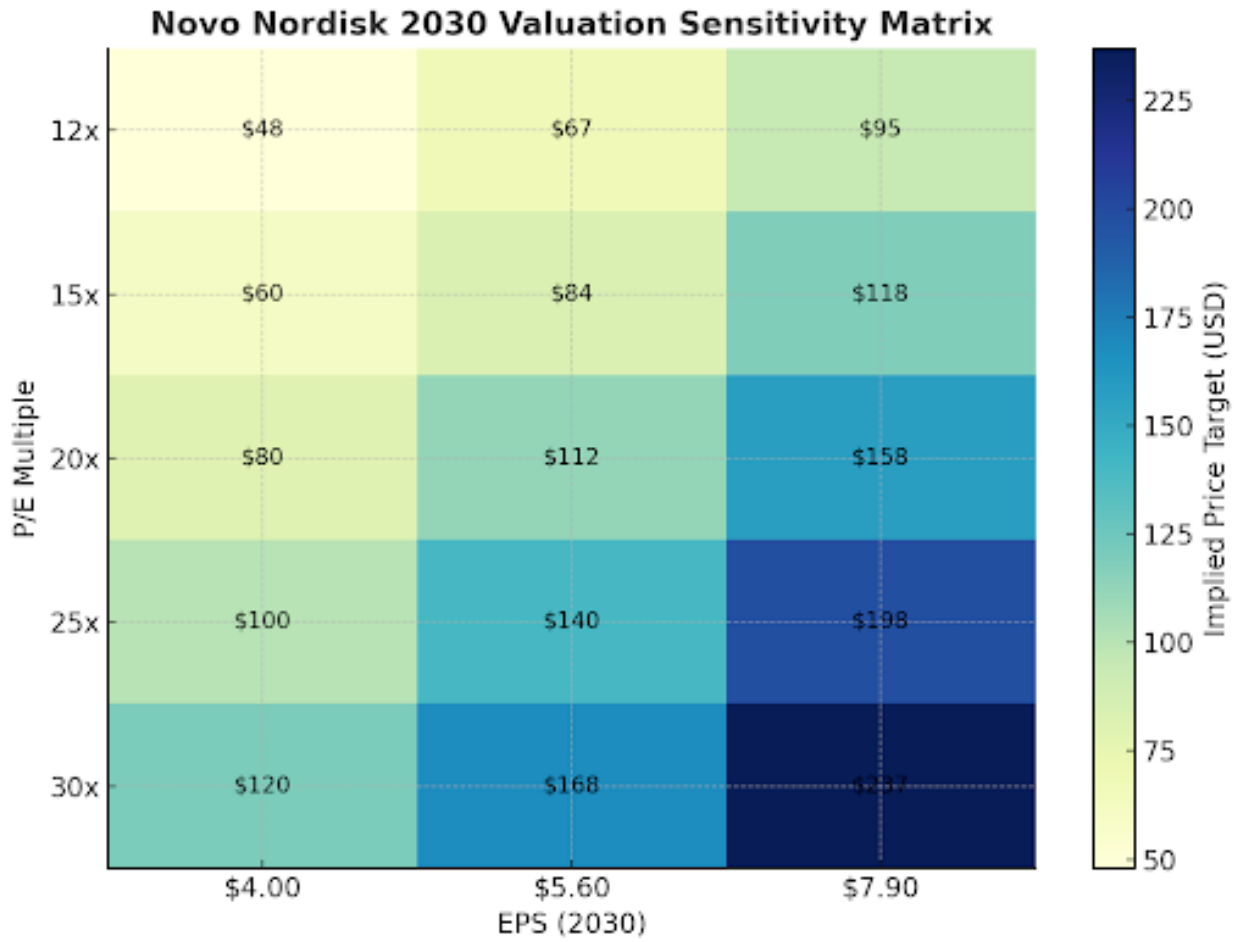
1.1. Novo Nordisk

Despite strong yet deceiving annual and quarterly performance, Novo Nordisk's share price has plummeted this past year, decreasing more than 55% year on year. The decrease was strong yet gradual, acting as an aggregation of negative signals. The primary reason for this drawdown is the lowering of the expectation of the present value of future cash flows, as investors had implicitly priced Novo Nordisk as the near-monopolist beneficiary of a multi-decade obesity-drug market. A "winner takes most" predisposition has transitioned into a contested market and Eli Lilly's operational success has proved to be the biggest cause for this revision, having reported $5.2bn of Mounjaro sales in Q2 2025 alone. Further negative signals sent by Novo cutting its 2025 sales growth guidance multiple times as management revised expected Wegovy momentum lower, as well as the appointment of a new CEO, put further doubt on the firm's competitiveness. Furthermore, Novo repeatedly cited competition from compounded or copycat versions of Wegovy as a reason sales growth was weaker than expected.

Novo Nordisk was surely overvalued a year ago, as truly unrealistic assumptions had been priced into the company's share price. However, there is a strong case for optimism for potential long-term investors in the company. The company has sent the message that it was ready to focus on efficiency in order to maintain its strong margins while pricing itself more competitively, focusing on speed and execution in its leading markets. On September 10th 2025, the company announced 9,000 layoffs. The move is expected to deliver cost savings of 8 billion Danish crowns (USD $1.25bn) annually by 2026, and the market reacted with relative optimism. Novo Nordisk conserves good fundamentals: a well-established manufacturing base, entrenched global sales channels, and a strong balance sheet, with operating margins well above 30%. A bullish view can be further backed by a P/E ratio of 14.27 and the premise that ultra-large capitalization equities have an increased capacity to bounce back from large drawdowns in a "too big to fail" logic as they retain access to financing, state support, and long-term investor trust. Furthermore, the hypothesis that during distress episodes, noise trading, herding, and forced liquidations amplify declines beyond levels justified by fundamentals, creating systematic opportunities for contrarian entries can be proved to be an effective systematic investing strategy.

	Revenue	EBITDA	Net Income
Quarter	+25.03%	+31.66%	+20.68%
Annual	+12.93%	+10.07%	+32.19%

Table 1: Novo Nordisk financial growth rates.



This sensitivity matrix shows how different assumptions on GLP-1 adoption and operational success from Novo Nordisk might reflect on the company's share price. The sensitivity matrix reflects the fact Novo is currently priced at moderate levels, and the potential upside significantly exceeds the potential downside of a bear case scenario. The conclusion from this analysis is that Novo Nordisk can be a valuable addition to long-term investors' portfolios.

1.2. Eli Lilly & Co

While Novo Nordisk has been aggressively repricing, Eli Lilly has spent the past two years executing one of the most aggressive growth stories in large-cap pharma. Its GLP-1 franchise,

propelled by tirzepatide (marketed as Mounjaro for diabetes and Zepbound for obesity), has rapidly scaled. In Q2 2025 alone, Mounjaro generated $5.2 billion in sales, representing year-over-year growth above 80% and underscoring Lilly's ability to translate clinical superiority into commercial traction. Analysts now project Lilly could capture 40–45% of the global GLP-1 market by the end of the decade. Clinical data suggest that tirzepatide delivers superior weight loss outcomes versus semaglutide, with patients often achieving 20%+ reductions in body weight, compared to ~15% for Novo's Wegovy.

While Eli Lilly's strategic strengths are undeniable, with a P/E ratio of 47x, the company is priced more akin to a high-growth tech stock than a mature pharmaceutical company. The market is effectively discounting a near-monopolistic scenario in obesity care, sustained innovation, and durable exclusivity. While near-term growth is undeniable, such valuation leaves little margin for error. Any clinical setbacks, pricing reforms, or slower-than-expected uptake could disproportionately impact investor returns. Short-term stock price fluctuations are particularly vulnerable to clinical trial results. The firm's long-term prospects, however, seem safe due to the attraction of long-term institutional investors, the company's diversified product portfolio and confirmed operational efficiency.

	Revenue	EBITDA	Net Income
Quarter	+37.65%	+60.94%	+60.38%
Annual	+32%	+81.40%	+105.63%

Table 2: Eli Lilly financial growth rates.

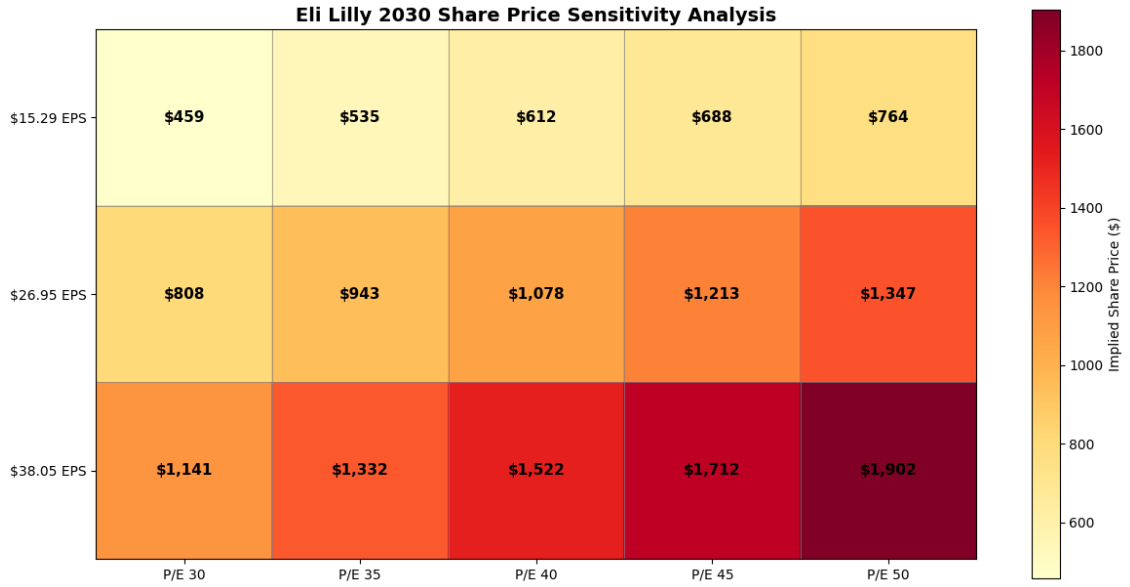


Figure 1.1: Eli Lilly &co. 2030 Valuation sensitivity matrix

This sensitivity matrix highlights how Eli Lilly's valuation is heavily predicated on continued clinical and commercial outperformance. The analysis shows that while upside potential

exists in a bull case of durable market leadership, the company's premium multiple means downside risks in the bear case are amplified. The conclusion from this analysis is that Eli Lilly remains a compelling growth story, with less uncertainty than Novo Nordisk, but at the price of a higher premium. The company is ultimately a valuable addition to long-term investors.

1.3. The threat of new entrants in expected duopoly dynamics

While Eli Lilly and Novo Nordisk currently dominate the GLP-1 landscape, history suggests that pharmaceutical duopolies rarely remain uncontested for decades. Several dynamics heighten the risk of disruption:

- Pipeline Competition. Amgen, Pfizer, and smaller biotech firms are developing oral GLP-1s and multi-pathway agents (GLP-1/GIP, GLP-1/GCGR, or even triple agonists). Oral delivery in particular could overturn the current injection-based paradigm by boosting compliance and broadening the addressable population. If efficacy proves comparable, adoption could accelerate rapidly.
- Patent Cliffs and Biosimilars. Both Novo and Lilly face eventual expirations on semaglutide and tirzepatide patents. Although biologics are complex to replicate, biosimilars (and compounded versions, already flagged by Novo as a pressure point) threaten to erode pricing power sooner than the patent timelines might imply.
- M&A Dynamics. Large-cap pharma players with thinning pipelines (e.g., Merck, Sanofi, AstraZeneca) are potential acquirers of emerging GLP-1 innovators. Such deals could provide new entrants with both capital and commercial scale, compressing the incumbents' market share, although their solid balance sheets could potentially allow them to do the same.
- Policy and Payer Pushback. Governments and insurers are unlikely to tolerate runaway drug spending indefinitely. Competitive bidding and formulary exclusivity could shift volumes away from incumbents if cheaper alternatives prove "good enough."

These potential factors are likely to alter the duopoly-like dynamics currently priced into the shares prices of Eli Lilly and Novo Nordisk. They remain the safest options for exposure to the obesity drug market, but any long-term positioning must factor in the asymmetric risk that new entrants, policy shifts, or patent erosion could compress margins and disrupt the duopoly.

1.4. Conclusion

The global obesity epidemic is reshaping healthcare and capital markets in tandem. Novo Nordisk and Eli Lilly stand as the clearest beneficiaries, having already established commercial scale and clinical credibility. Yet, their trajectories diverge: Novo, trading at more modest multiples after a steep correction, presents an opportunity for contrarian long-term investors willing to weather competitive pressures. Lilly, meanwhile, commands a premium valuation that leaves little room for execution missteps, but reflects undeniable clinical leadership.

For portfolio construction, investors should resist binary thinking, allocating exposure proportionally to conviction in each firm's execution, while acknowledging the disruptive potential of emerging entrants. Pipeline optionality remains a critical lens: beyond GLP-1 incumbents, dual and triple agonists, oral formulations, and novel mechanisms offer asymmetric upside for early-stage biotech investors. M&A dynamics are likely to accelerate, with smaller players becoming acquisition targets as big pharma seeks to deepen its metabolic portfolios. Meanwhile, second-order effects, ranging from supply chain bottlenecks to consumer behavior shifts, create investable themes in CDMOs, cold chain logistics, and wellness platforms.

Strategically, investors must balance conviction with caution. Thematic exposure via healthcare ETFs or metabolic-focused funds offers breadth, while targeted positions in pipeline-rich

firms provide depth. Whatever their convictions and risk tolerance, tracking changes in the obesity drug market will provide investors with great opportunities moving forward.

1.5. Trading application: Directional relative-value pair: Long NVO / Short LLY

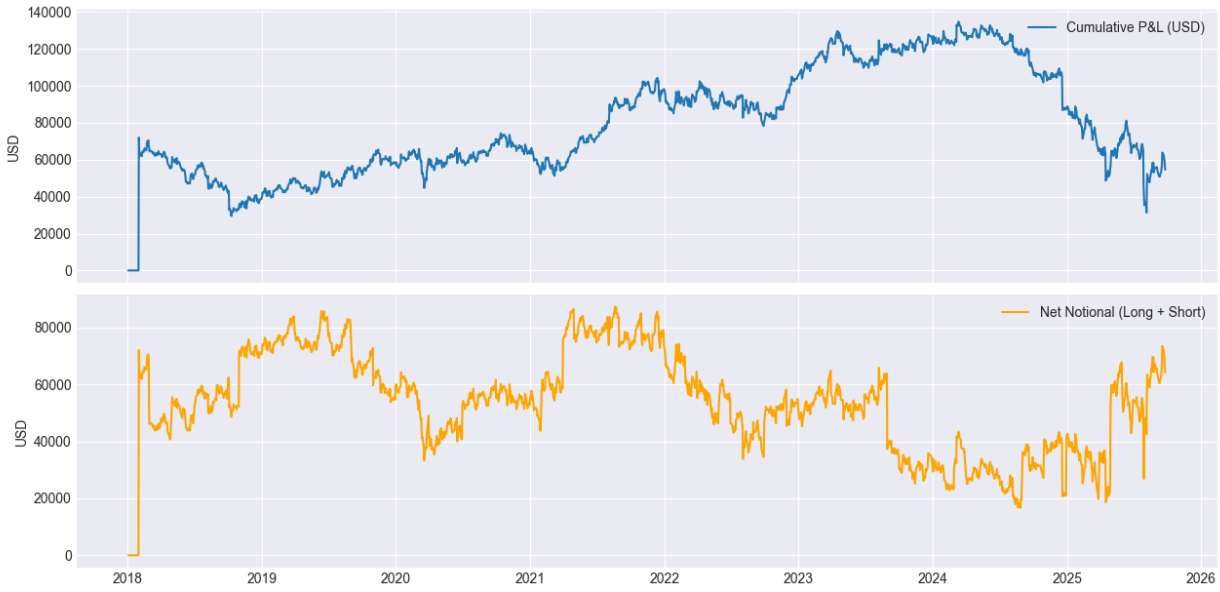
A long-term, directional pairs-trading strategy can be an intuitive implementation of a strategy focusing on mean-reversion in the NVO-LLY pair. Based on the conviction that Novo Nordisk will revert back to a higher relative price ratio with Eli Lilly, on the premise that Eli Lilly's share price has grown beyond a fair valuation and Novo Nordisk's share price has fallen below a fair valuation. The main justification for such a trade would be the market's "winner-takes-most" bias toward Lilly is likely overextended and Novo's fundamentals, cost-cutting measures, and efficiency initiatives provide a margin of safety.

The strategy, structured as a long NVO/short LLY bet to capitalize on a reversion in their valuation gap, was tested across three scenarios: a baseline with no risk controls, a version with a 20% hard stop-loss, and a version with a 15% trailing stop. The original, unmanaged strategy delivered a seemingly respectable 54% return. However, this return was achieved only by surviving a harrowing -68.2% maximum drawdown, a loss magnitude that would trigger liquidation for most professional funds and test the resolve of any investor. The low Sharpe ratio of 0.19 confirms that these returns were achieved with a mostly unacceptable level of risk.

Strategy Variant	**Return (%)**	**Max Drawdown (%)**	**Sharpe**	**Volatility (%)**	**Days Active**
No Stop-Loss	54.3	–68.2	0.19	38.5	1,926
Hard Stop –20%	52.1	–18.7	0.27	26.3	88
Trailing Stop –15%	56.8	–14.7	0.29	26.2	58

Table 3: Performance comparison of strategy variants.

The insight of the strategy lies in the timing of the stop-loss triggers. Both the hard stop and the trailing stop were activated within the first three months of the strategy, on April 25 and June 7 of 2018, respectively. The market was sending a clear, early signal: the fundamental relationship underpinning the trade had broken down. The original strategy, by ignoring this signal, remained invested for over five additional years, holding on through a period of extreme volatility.



The first important point to make is that the stocks are not cointegrated at 5% significance level (p-value=0.9708) . Past cointegration, prior to large divergences fuelled by differences in operational execution and trend expectation, did not carry for long and made such a strategy unfeasible when accounted for risk. Without cointegration, the reversion thesis is closer to a directional bet on relative valuation rather than a statistical arbitrage strategy.

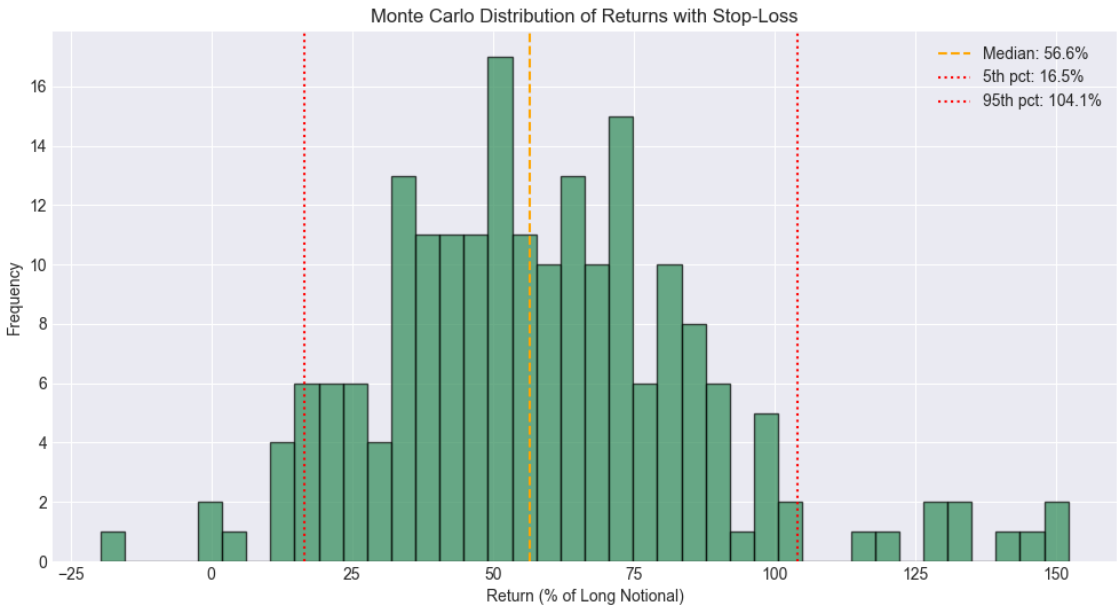
Two complementary methods were applied to simulate future results of this directional strategy: Bootstrap resampling and Monte Carlo simulations of price paths.

Percentile	**Total Return (%)**
1%	1.11
5%	16.54
10%	23.41
25%	39.07
50% (Median)	56.60
75%	74.03
90%	90.31
95%	104.12
99%	144.44

Table 4: Price-Level Monte Carlo – 200 simulations (with Stop-Loss), showing percentiles of total return on long notional.

Monte Carlo Simulation creates hypothetical forward price paths by drawing from a statistical model of returns, typically using the estimated means, volatilities, and correlations between Novo Nordisk and Eli Lilly. Each simulated path is converted into price series. The

full trading strategy, including rebalancing rules, transaction costs, and stop-loss mechanics is then run on that synthetic history. The result is a distribution of possible outcomes, not limited to what has been observed historically, but instead extrapolated under the assumption that the estimated return process remains stable. The simulations incorporated risk management through stop-loss mechanisms that bounded extreme drawdowns.



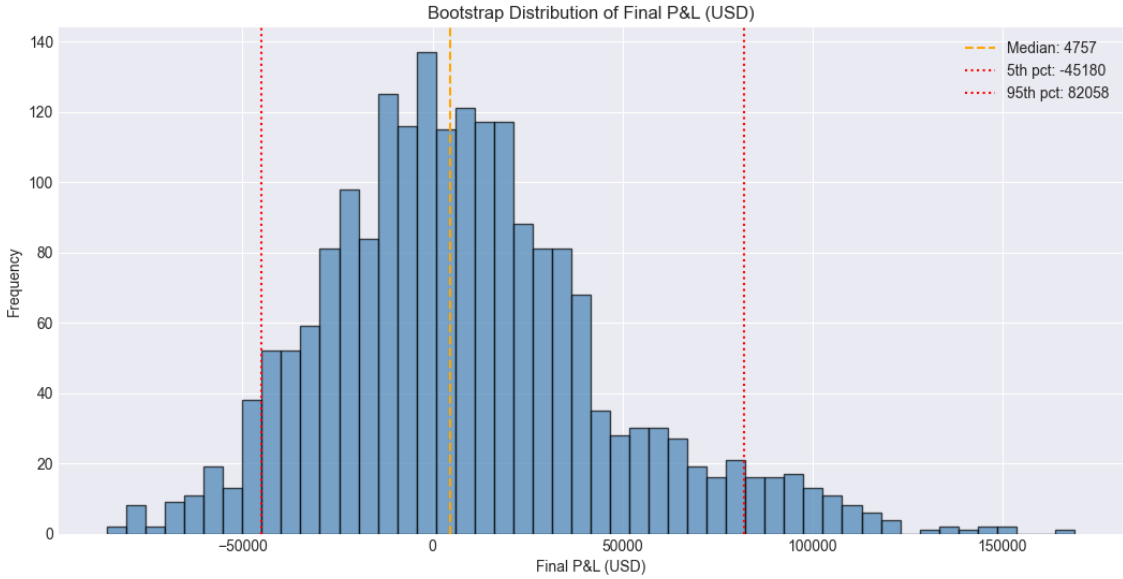
Stop-loss mechanisms offer an effective safeguard in periods where the mean reversion thesis fails. It allows possible execution to benefit from the high returns offered in the right-tail of the strategy distribution while saving it from the sharp losses incurred in the left-tail scenarios.

Percentile	**P&L (USD)**
1%	–68,834.56
5%	–47,201.28
10%	–37,108.38
25%	–17,580.36
50% (Median)	3,915.71
75%	25,250.30
90%	51,207.15
95%	73,784.99
99%	106,633.35

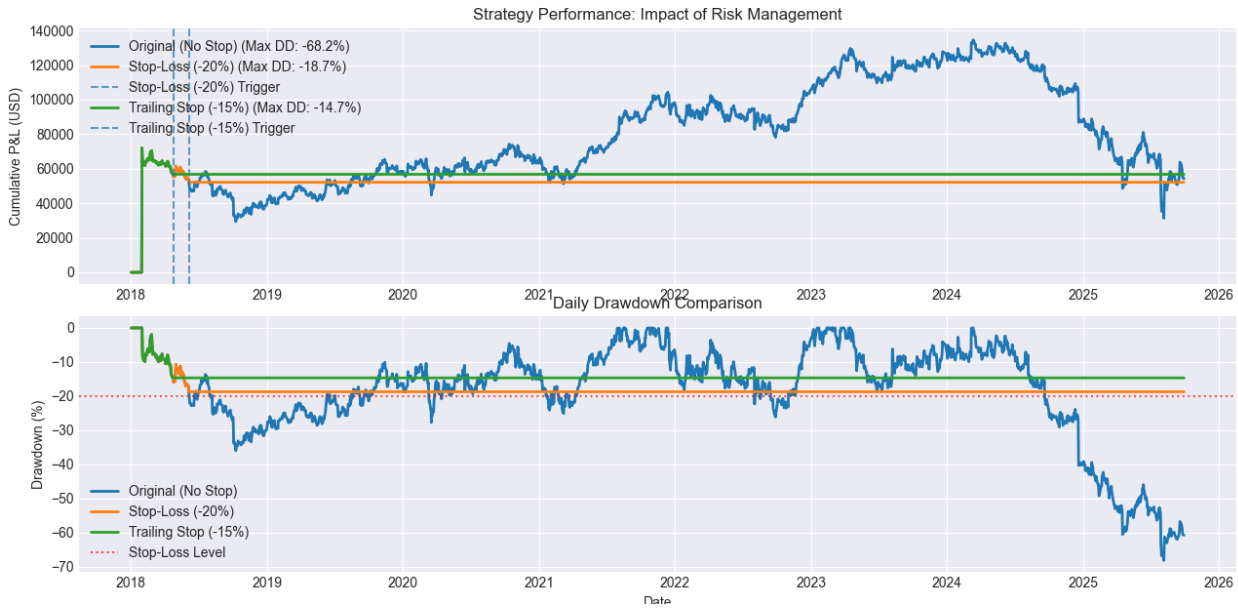
Table 5: Bootstrap Resampling – 200 simulations: Final P&L distribution (USD).

Contrarily to Monte Carlo simulations, bootstrap resampling takes the historical sequence

of daily strategy P&L and resamples it in blocks to generate synthetic paths. It analyses the impact alternative dynamics in the chosen pairs's stock prices might have had. Bootstrap resampling does not impose past return dynamics on the future, which is crucial for pairs-trading or mean-reversion strategies, where assuming cointegration or autocorrelation structures from history could bias risk assessment. Instead, it explores the range of outcomes that are consistent with the actual distribution of past strategy performance.



Taken together, these simulations illustrate the fragility and optionality of the NVO/LLY directional trade. The trade offers explosive upside under favorable valuation reversions, but its risk profile is highly asymmetric where modest deviations from the mean-reversion thesis produce large losses. Depending on the convictions and risk tolerance of possible implementers, the strategy can be optimized into being acceptable, despite both stocks lacking cointegration. An inevitable factor to consider for any viable execution is the implementation of risk management, possibly through stop-loss or trailing-stop mechanisms when the premise of the strategy breaks down.



In this analysis, the rebalancing was done on a monthly basis and the strategy was beta-neutral. While this article predicts growth in the obesity-drug market, the beta-neutral execution was used to purely analyse the potential for NVO/ LLY dynamics. The factors presented in this article imply the the obesity-drug market is set to grow in the coming years: a long-biased strategy would take this into consideration in the strategy. The expected returns for a long-biased strategy can be expected as being in the following range:

Scenario	Monte Carlo (Beta)	Bootstrap (Resampling)
1.5x Baseline Long, 70% Beta		
1st pct	14.66	-66679.04
5th pct	18.13	-47583.06
10th pct	22.46	-36679.44
25th pct	41.00	-17290.33
50th pct	78.85	5330.48
75th pct	83.40	28156.52
90th pct	90.76	55905.92
95th pct	102.27	75240.10
99th pct	111.48	115372.56
2x Baseline Long, 70% Beta		
1st pct	14.78	-73629.75
5th pct	18.25	-48027.61
10th pct	22.58	-35809.28
25th pct	41.13	-16055.16
50th pct	78.96	5629.17
75th pct	83.56	26329.21
90th pct	90.90	54450.00
95th pct	102.41	73119.18
99th pct	111.62	119288.66
2x Baseline Beta, 100% Beta		
1st pct	14.78	-69307.77
5th pct	18.25	-46227.33
10th pct	22.58	-35806.17
25th pct	41.13	-16092.40
50th pct	78.96	3864.67
75th pct	83.56	27885.69
90th pct	90.90	55856.04
95th pct	102.41	75785.79
99th pct	111.62	105060.79

Table 6: Beta Hedge Monte Carlo and Bootstrap Percentiles

When extending the analysis to long-biased variants, the distribution of outcomes shifts meaningfully. By scaling the long side relative to the short hedge—whether through a 1.5× or 2× notional tilt, or by relaxing the hedge to 70% of calculated beta—the simulations consistently show stronger right-tail performance, with medians approaching +79% and 90th percentiles above +90%. However, bootstrap resampling again tempers this optimism: the 1st percentile remains severely negative (–$66k to –$73k), illustrating that leverage on the long leg magnifies both upside optionality and left-tail fragility.

The key takeaway is that risk-managed, long-biased strategies can offer an attractive convex profile: exposure to Novo Nordisk's structural growth story with protection against catastrophic divergence. Yet, the sustainability of such trades depends critically on the timing of

stop-loss triggers and the persistence of fundamental drivers. In an industry as momentum-driven as obesity drugs, where market leadership can abruptly shift, the statistical structures assumed by Monte Carlo can quickly become obsolete.

1.6. Assumptions made in Valuation sensitivity matrixes

- Bear case scenario: 0% CAGR
- Base scenario: 12% CAGR
- Bull case scenario: 20% CAGR
- Shares outstanding assumed constant
- EPS 2024 used as starting point (NVO = 3.95, LLY = 15.29)
- P/E ranges applied: 12–30x for NVO, 30–50x for LLY
- Market size assumption: GLP-1 market grows to $100–120 billion by 2030
- Novo Nordisk: operating margin assumed: 35% (in line with historical 2023–2024 performance, which was typically above 30%).
- Eli Lilly: operating margin assumed: 40% (reflects its current high profitability from GLP-1 growth and broader portfolio leverage, in line with historical performance).
- Tax rate: 18% (consistent with reported effective tax rate in recent years).